Lock-In AI Meal Planner MVP Report (Team 10)

Group Members: Ryan Craft (rcraft13@asu.edu), Stanley Drake De Castro (shdecas1@asu.edu), Devin Gallegos (dgalle14@asu.edu), Wells Marcus (wbmarcus@asu.edu), and Carter Roane (chroane@asu.edu)

1. Executive Summary:

Lock_In is an AI-assisted meal planning application that helps users generate personalized meal plans based on their goals, preferences, allergies, and budget. The main goal of the project is to make meal planning easier and more realistic by handling multiple constraints at the same time. In phase 3, we expanded our Phase 2 prototype into a more complete MVP. The system now supports multi-day meal planning, grocery cost estimation, and user interaction for adjusting plans. While the current version is still mostly rules based, it does a much better job handling real constraints and sets up a strong foundation for future AI integration.

2. User & Use Case:

The main users are people who want to eat healthier but don't want to spend a lot of time planning meals. Many people struggle to balance things like calories, protein, budget, allergies, and preferences all at once.

A typical use case is a user entering their information into the web app, such as their dietary goals, preferences, and constraints. The system then generates a multi-day meal plan along with nutrition targets and an estimated grocery list. The user can also adjust the plan as needed.

3. System Design:

The system is built with a modular structure that separates the frontend, backend, and planning logic.

The frontend is a simple browser-based interface (static/index.html) where users input their information and view results. The backend is a Python server (app.py , server.py) that handles API requests and sends responses. The core planning logic is implemented in planner.py, with additional modules supporting pricing, nutrition lookup, and user interaction.

The system includes API endpoints such as /api/plan for generating meal plans, /api/interact for modifying plans, and /api/cheaper-plan for reducing cost.
Data flows from the frontend to the backend through API requests. The backend then calls the planner logic to generate a meal plan, and the results are sent back to the frontend to be displayed to the user.

4. Data:

The system uses several structured data sources:

- meals.json for meal and ingredient information
- prices.json for estimating grocery costs
- fdc_cache.json for nutrition values
- raw_data/ for original dataset files

The data is cleaned and organized so the planner can efficiently generate and evaluate meal combinations. Since the system is rules based and does not use a trained model, a train/test split is not required.

5. Technical Approach:

The current MVP uses a rules based approach instead of a trained machine learning model.
The planner selects meals based on user inputs and filters them using constraints based on how well they match calorie goals, variety, and other factors.
The system also:

- estimates grocery cost using pricing data
- attempts to generate cheaper versions of meal plans
- allows users to interact and refine plans

There is a placeholder for OpenAI integration (openai_client.py), but it is not fully implemented yet. The system is designed so that AI can be added later to improve personalization and decision making.

6. Evaluation:

The system was evaluated based on whether it works end to end and produces reasonable outputs.

The application successfully:

- takes user input through the frontend
- generates multi-day meal plans
- calculates nutrition targets
- estimates grocery costs
- allows users to modify and refine plans

For example, a user can input a calorie goal and budget, and the system generates a weekly meal plan along with a grocery estimate. This demonstrates a complete end to end workflow from user input to final output.

Unit tests in the tests/ folder help verify that key components such as the planner, pricing logic, and nutrition lookup are working correctly.

7. Limitations & Risks:

Limitations of the system include:

- Meal data is limited and may not cover all diets
- Cost estimates are based on static pricing
- There is no user account system or saved data
- Constraint handling is not perfect and may not always find the best plan

The largest risk would be the system making a mistake and recommending a recipe the user is allergic to. Users may also assume the results are fully accurate nutrition advice, even though the system is not designed for medical use. There may also be a risk of privacy since users input personal information.

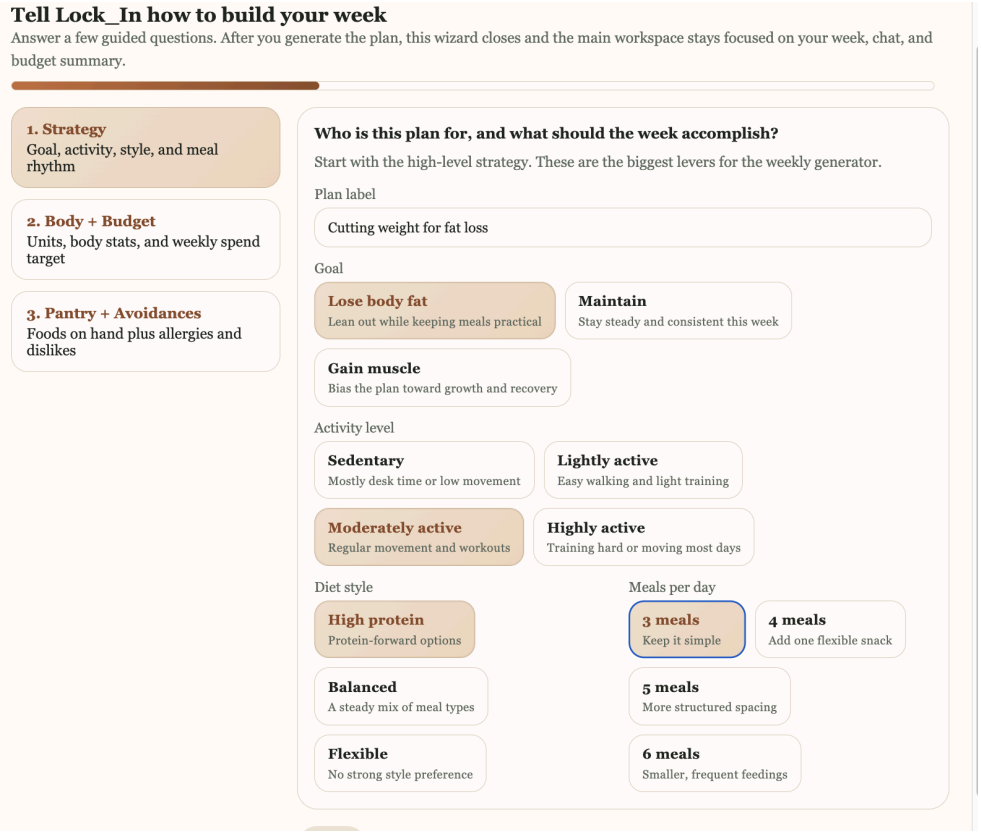
8. Next Steps:

With more time we would:

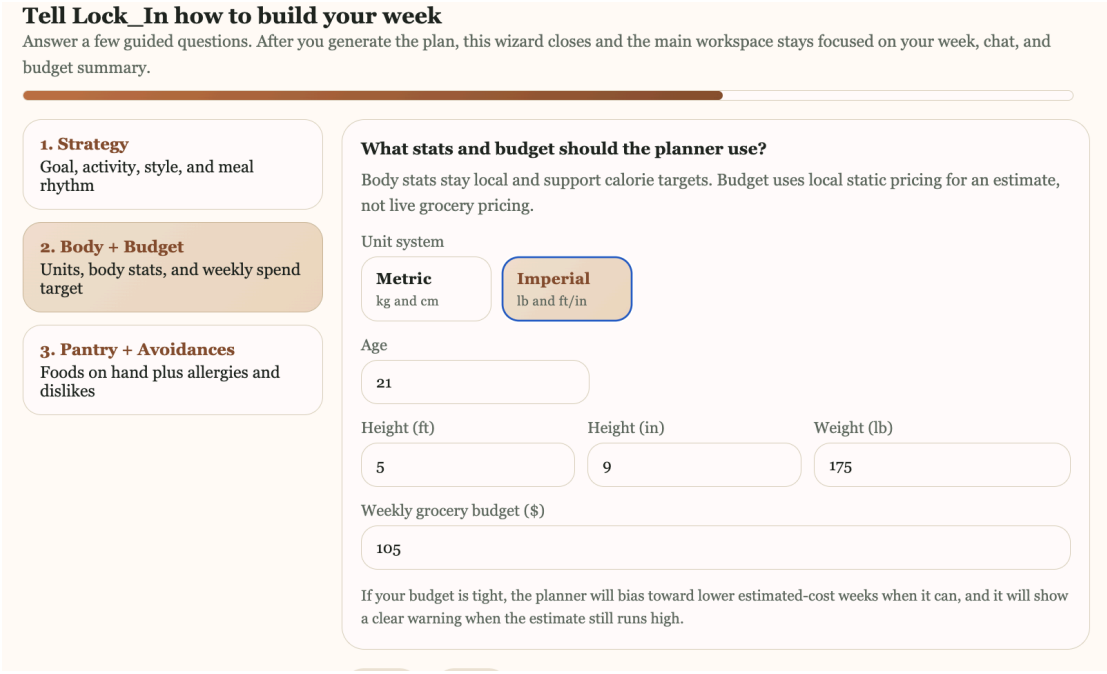
- integrate OpenAI or another LLM to improve meal generation and personalization
- improve constraint checking to ensure plans are always valid

- expand the meal dataset and improve ingredient tagging
- add user accounts and saved plans
- improve the frontend design
- improve cost and nutrition optimization

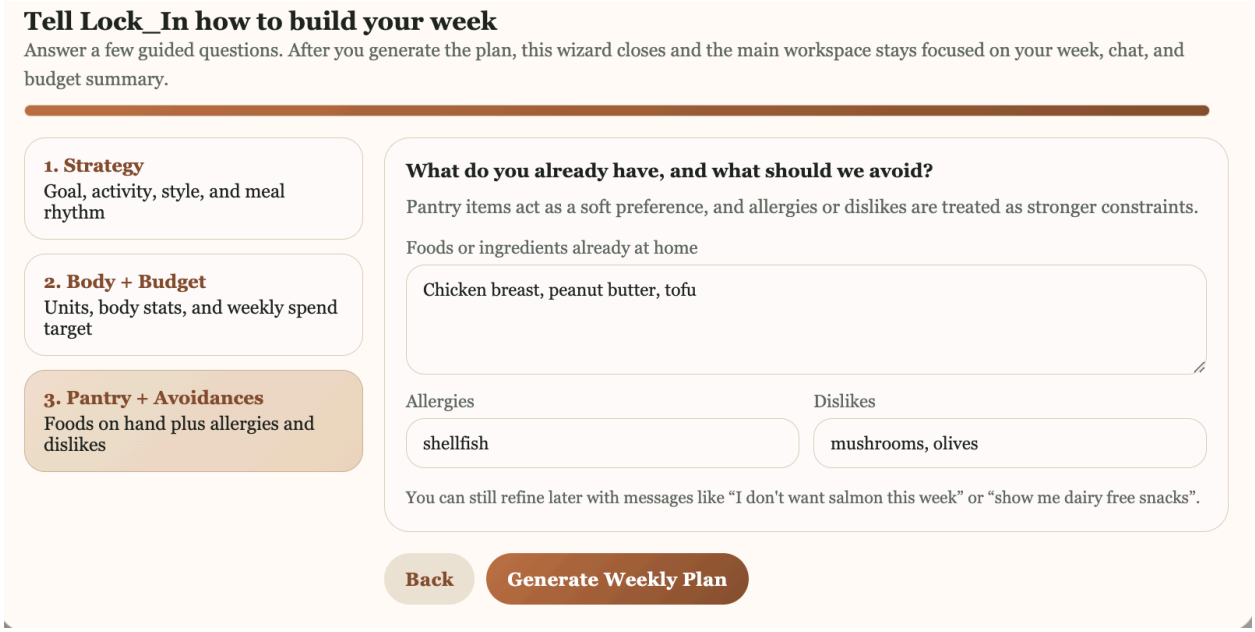
How It Actually Works



To start, you are asked what your general goals are for your lifestyle. Things such as weight management, activity level, dieting style, how often you want to eat, etc.



Next you are asked about more personal questions to help tailor the system's output. This includes things like age, height, weight, and weekly budget to spend on food.



To tailor the output even further, you are then asked for ingredients you have on hand already, any allergies you may have, and even preferences to foods you do not enjoy. This helps to avoid creating any meals that you are not able to eat or not able to

LOCAL WEEKLY PLANNER

Build the week, then keep shaping it.

The main workspace stays focused on your weekly meal plan, budget snapshot, and natural-language refinements. Reopen setup any time if your goals, budget, pantry, or units change.

Edit Preferences

PROFILE SNAPSHOT

Goal	**Fat Loss**
Activity	**Moderate**
Diet style	**High Protein**
Meals per day	**3**
Units	**Metric**
Pantry / avoids	**3 pantry items · shellfish / mushrooms, olives**

BUDGET SUMMARY

Weekly cost snapshot

Budget is estimated from local static ingredient pricing. Pantry items are considered as soft planning hints.

Estimated grocery spend	**$281.40**
Budget target	**$105.00**
Pantry items considered	**3**
Budget status	**Over by $176.40**

GROCERY LIST

Review your shopping list

Open the grocery workflow to review the combined list, mark what you already have, and apply pantry updates for this session.

WEEKLY PLAN

Your generated week

The plan stays compact by default. Open any meal to see ingredients, directions, tags, allergens, notes, and nutrition.

Cutting weight for fat loss's Weekly Plan

This weekly plan was generated by Lock_In's candidate-and-scoring engine to target about 2373 kcal/day and 159g protein/day for fat loss with high protein preferences. It selected from multiple weekly candidates, favoring variety and macro fit. The chosen plan uses 21 unique meals and averages about 1982 kcal/day. It also biased toward meals that use your available foods and found about 16 ingredient overlaps across the week. The consolidated grocery estimate is roughly $281.40 for the week against your $105.00 target, with about 30% ingredient price coverage from local static pricing.

Budget target: $105.00 · Estimated grocery spend: $281.40 · Status: Estimated Over Budget

Weekly budget is estimated from local static ingredient pricing, so treat it as directional rather than precise grocery pricing. The current weekly estimate is about $281.40, which is above the $105.00 budget target. Some meals relied on category fallback pricing because not every ingredient had a direct local price match. Inventory matching is best-effort and currently looks for ingredient overlap in meal names and ingredient lists. Pantry-aware planning found about 16 ingredient overlaps and biased toward meals that use those foods.

1982 avg kcal/day · 2373 kcal/day target · 119g avg protein · 286g carbs · 66g fats · 2.8L water · ~$281.40 grocery spend

Monday
2042 kcal across the displayed meals

- Meal 1 · Breakfast **Fried Cornmeal Shrimp with Butternut Squash Risotto** — 27g protein
- Meal 2 · Lunch **Cornmeal-Crusted Chicken with Toasted Corn Salsa** — 44g protein
- Meal 3 · Dinner **Spiced Chicken Couscous with Cinnamon Onions** — 18g protein

Tuesday
2277 kcal across the displayed meals

- Meal 1 · Breakfast **Grilled Cheese with Smoked Turkey and Avocado** — 35g protein
- Meal 2 · Lunch **Quick Sesame Chicken With Broccoli** — 56g protein
- Meal 3 · Dinner **Classic Chicken and Coconut Soup** — 41g protein

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Need to buy	**113 consolidated items**
Estimated grocery spend	**$281.40**
Estimate type	**Approximate static pricing**

Alfalfa Sprout — Top remaining item · 1 meal uses — Over budget

View Grocery List

Wednesday
2056 kcal across the displayed meals

- Meal 1 · Breakfast **Cobb Salad Pitas** — 34g protein
- Meal 2 · Lunch **Chicken Tagine with Chickpeas and Mint** — 52g protein
- Meal 3 · Dinner **Chicken and Corn Stew with Corn Wafers** — 57g protein

Thursday
1702 kcal across the displayed meals

- Meal 1 · Breakfast **Hoisin Chicken in Lettuce Leaves** — 29g protein
- Meal 2 · Lunch **Grilled Chicken Breasts with North African Spice Paste** — 58g protein
- Meal 3 · Dinner **Coriander Chicken Tostadas with Refried Beans and Grilled Fennel** — 46g protein

Friday
1894 kcal across the displayed meals

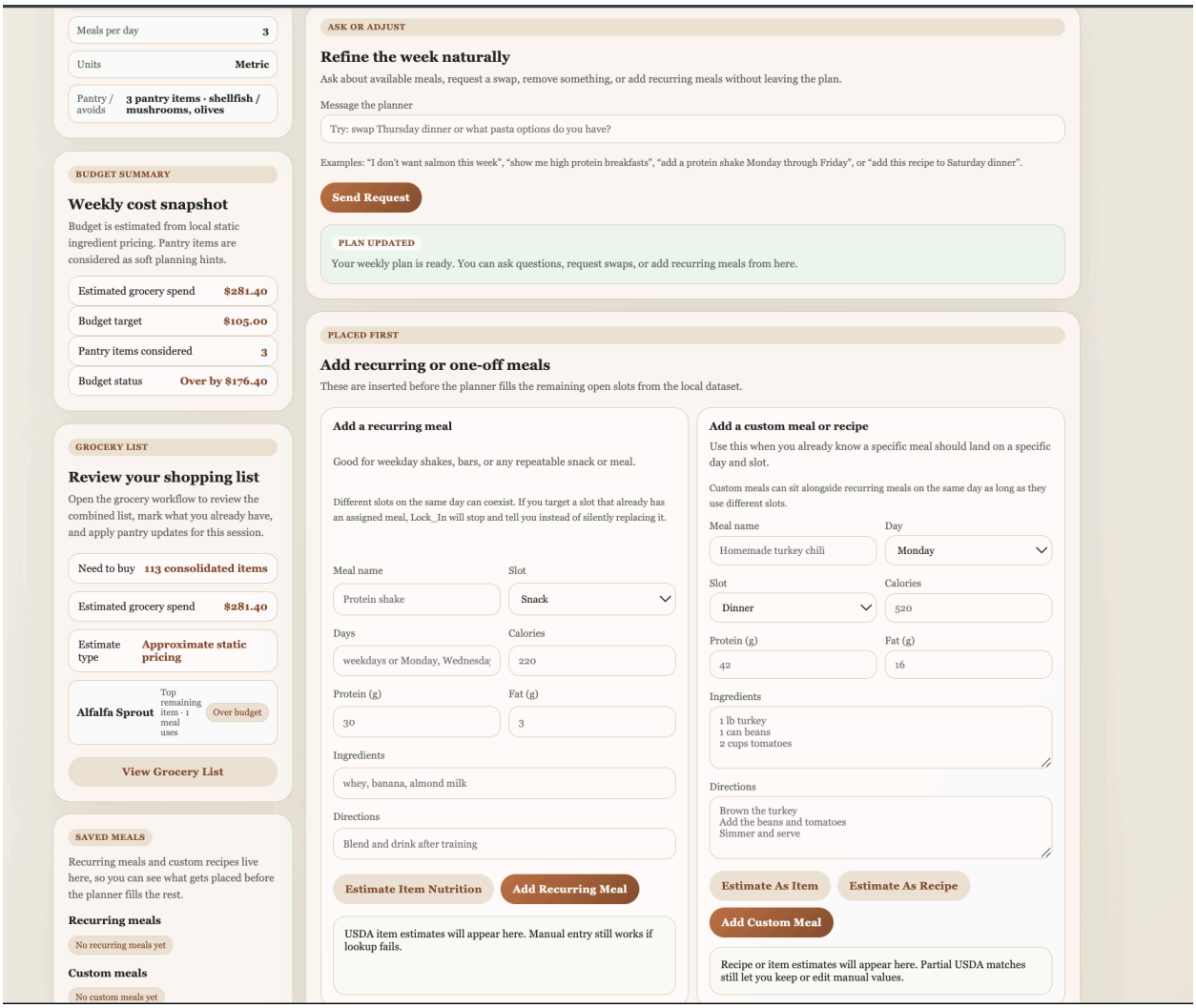
- Meal 1 · Breakfast **Peanut Butter, Banana, and Jelly "Ice Cream"** — 4g protein
- Meal 2 · Lunch **New Chicken Parmesan** — 49g protein
- Meal 3 · Dinner **Peanut Butter, Apple, and Bacon Sandwiches** — 39g protein

Saturday
2318 kcal across the displayed meals

- Meal 1 · Breakfast **Fried Rice with Bacon, Scallions, and Water Chestnuts** — 15g protein
- Meal 2 · Lunch **Herb-Roasted Chicken Breasts** — 53g protein
- Meal 3 · Dinner **Walnut Chicken Salad with Green Apple Vinaigrette** — 97g protein

Sunday
1588 kcal across the displayed meals

- Meal 1 · Breakfast **Grilled Avocado with Strawberry-Mango Salsa** — 5g protein
- Meal 2 · Lunch **Curried Chicken Salad** — 29g protein
- Meal 3 · Dinner **Chicken Salad with Lentils, Peas and Mint** — 48g protein



And finally, after all of the personal questions have been answered, the system will output a set weekly meals to follow to meet the goals of each user. The very last screenshot shows post-generated diet features, one of which allows you to refine/swap your diet if there ends up being a meal that you realize you actually do not want to eat that week. The system will alter its plan on the spot to fit your wants and needs. On top of this, if there is a recurring meal you plan on having multiple times a week like a protein shake or a specific snack, the system can be updated to accommodate these wants and alter the provided diet to fit any input within our datasets. And the final feature is being able to add your own custom meal. If there is something specific you know you may want to eat at a specific time, you can input it into the system and have the plan regenerated to fit the exact specifications you may have. This is especially helpful

if you end up not liking any of the recommendations made from the refine/swap option, it allows you to take matters into your own hands and control that specific scenario.

Note: One of the only problems we encountered was pricing, if someone has quite a low amount of money to work with, sometimes the system will somewhat ignore it and provide a diet that is not close to within the users budget. This can be seen above where the budget is $105, but the estimated price is $281.40. We were not able to fully work this out in time, but we believe it to be related to the system not having things like name brand vs store brand implementations. These are all from static prices, making it difficult to fit into tighter budgets.